FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling private-sector-led solar-powered Irrigation Systems (SPIS) for climate-resilience in smallholder farming in Sub-Saharan Africa

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 1 · Close the Climate Adaptation Gap through Scalable Climate Services and Smart Water Management

Flagship 1.2 · Scaling Smart Irrigation & Climate-Resilient Water Systems

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management
  - **Flagship 1.1** — Digital Climate Services for Resilient Farm & Enterprise Management
  - **Flagship 1.2** — Scaling Smart Irrigation & Climate-Resilient Water Systems
- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems
  - **Flagship 2.1** — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems
  - **Flagship 2.2** — Entrepreneurship through Mechanization & Postharvest Services
  - **Flagship 2.3** — Genetics, Inputs & Services for Animals & Aquaculture
- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition
  - **Flagship 3.1** — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

- **Description of the innovation package**

Irrigation is a critical lever for climate resilience, yet coverage remains insufficient: less than 10% of farmland in sub-Saharan Africa is irrigated. Existing irrigation systems are often inefficient and water-intensive. While climate-smart irrigation technologies such as solar-powered irrigation pumps (and farmer-led irrigation, such as fuel-efficient pumps) are proven at scale and ready for adoption, adoption remains fragmented due to high upfront costs, limited finance, weak supply chains, and inadequate advisory support.

The core innovation of this scaling track is mainly **a bundled solar-powered irrigation system (SPIS) scaling package** consisting of solar-powered irrigation pumps (validated, market-ready), financial innovations to de-risk investments using public, and financial institutions' funds, and digital advisory to improve SPIS adoption by smallholders and improve FLID irrigation performance through strengthened extension systems and policy support to enhance water system resilience to climate change.

While PayGo/PayOwn bundling with solar-based irrigation is central to the Inclusive Scaling Pathway of SPSI (eg, for Ghana , this track remains flexible and responsive to country-specific conditions. It actively explores complementary financial instruments aligned with local demand patterns, risk profiles, and technology supply chains. This ensures that financing solutions remain relevant and scalable across diverse institutional and market environments.

SPIS is already scaling-ready (level 9) and validated in real-world conditions, the PayGo as Sustainable Finance for Off-Grid Solar Irrigation: IPSR Innovation Profile (level 7), and Solar Irrigation Pump (SIP) Sizing Tool: IPSR Innovation Profile (level 7). The track, however, goes beyond SPIS to promote farmer-led irrigation (eg, efficient fuel pumps, hybrid systems, and improved practices) that enhance water productivity and system efficiency. S4I leverages the Solar phase II bilateral project to assess the suitability of SPIS and pump sizing in the two target countries, and the Gates Foundation initiative for irrigation mapping in three countries.

Evidence from Ghana showed that a private supplier increased its sales of solar-powered irrigation pumps up to 80% in 2021 from the market linkage pathway. The estimated annual investment in 2022 from private-sector partners is US$500,000, of which US$250,000 is allocated to the smallholder farmer segment. In Nigeria, Ethiopia, and Kenya FLID, mainly SPIS, is financially compelling when properly packaged and has an opportunity to scale. In Nigeria, IWMI co-designed a scaling pathway for SPIS to be farmer-owned, as investments in crops such as tomatoes, red peppers, onions, and rice have yielded very high returns. SPIS has also co-designed a scaling pathway with the Bank of Agriculture (BoA). In Ethiopia, SPIS is officially tax-exempt, benefit–cost ratios exceed one (). In Kenya, IWMI and SunCulture co-designed a water strategy that aligns SunCulture's solar-scaling operations with environmental sustainability goals. The intended package reduces investment risks, lowers upfront costs, and ensures sustainable groundwater use and higher returns.

- **Scaling ambition (2026–2028)**

By the end of 2026, our target is to reach 140,000 farmers (Table 1) through engaging with relevant intermediary actors along the irrigation supply chain, including private technology suppliers, financial institutions, development partners, and government agencies (scaling partners). The project also bridges gaps by facilitating investment in new solar technology units, enhancing the utilization of existing underutilized infrastructure through policy influence (in collaboration with climate action and policy innovation), digital tools, capacity strengthening for increased efficiency, and leveraging technical support to enable an environment through AoW3 of S4I. Solar irrigation scaling initiative adopts a common strategic approach across the three countries while adapting implementation pathways to national institutional systems, policy priorities, and market conditions. The project focuses on three core scaling pillars: technical and extension strengthening; financial innovation; and market linkage (e.g., farmers' awareness of SPIS, irrigation technology suppliers, and financial services providers) and advisory services, supported by a coordination platform, policy engagement, and knowledge generation. We aim for 2026:

- 23,500 farmers reached through market linkage and digital advisory for the adoption of SPIS by public and private scaling partners, so that they would be able to irrigate about 25000 ha,
- Application of finance facility to de-risk investment in three countries and policy support, which is projected to directly benefit 13000 smallholder farmers who invest in SPIS.
- Mobilizing 30 % private sector finance in support of the de-risking investment of scaling partners' investment
- Provide technical and extension strengthening support through 3 Training of Trainers (ToT) to technology service providers, FLID extension services providers, and media outreach, which is projected to directly benefit 105000 smallholder farmers who invest in SPIS and FLID.
- 1 policy brief in Ethiopia and 1 policy brief in Nigeria (in collaboration with Climate Action and policy innovation) that impacts Solar and FLID investments and sustainability in 2027 and 2028

Table 1: Country breakdown (based on partner capacity and pipelines) for 2026, 2027, and 2028

	2026	2026	2026	2027	2028
Farmers reached by	Ethiopia	Nigeria	Kenya	All	
Technical and extension support	50000	20000	35000	108000	122400
Market linkage and advisory services (including SPSI deployment)	9420	9050	5000	25548	27703
Finance access and policy support	300	5000	2000	71999	75239
Total	52800	39050	42000		
	140,770	140,770	140,770	205,547	225,342

By 2028, we aim to reach a cumulative total of 570,000 farmers. Through technical and extension support (such as ToT and implementation of curriculum developed under SolAR), it is projected to directly benefit more than 335,000 smallholder farmers by raising awareness of SPIS, and efficient FLID water use. Through market linkage and advisory services (such as demand-supply linkage, co-development of business models, and the application of digital advisory services for pump sizing), more than 75,000 smallholder farmers benefit from adopting SPIS through public and private institutions. Through scaling of finance facilities (such as application and scale of water tariff and blended finance in Kenya, and the application of identified investment ready supplier pipelines from AoW3 of S4I) and implementation of policy (such as implementation of PPP strategy in Ethiopia and scaling barriers in Nigeria from S4I AoW3), more than 150, 000 smallholder farmers benefit from new financial facility, PPP strategy and policy implementation.

Key demand and scaling partners

Demand for SPIS and FLID comes from governments seeking a PPP strategy (e.g., by *MoA in Ethiopia*), climate-resilient irrigation pathways (e.g., NEMA in Kenya), scalable markets by private suppliers (e.g., SunCulture in Kenya), and de-risked investment opportunities by financial institutions (e.g., NIRSAL and BoA in Nigeria). We identified key partners with whom we will work to achieve the targets. S4I does not finance the implementation of SPIS or FLID, but it allocates some funds to catalyze scaling, such as creating coordinated engagements for collaboration and training, providing one-to-one technical assistance, and demand-supply linkage. Key scaling partners were identified and selected because they already have national initiatives, have worked with IWMI and IFPRI, have demonstrated reach and credibility, and were active in the 2025 S4I activities.

In Nigeria, the Ministry of Agriculture and Food Security (*MoU with IFPRI*) serves as the policy anchor; National Science and Engineering Infrastructure (NASENI) and Nigeria Incentive-based Risk Sharing System for Agriculture (NIRSAL) lead market-based delivery of solar solutions; the Bank of Agriculture (MoU with IWMI) and Jaiz Bank provide access to finance; Ahmadu Bello University and Aliko Dangote University of Science and Technology support technical validation and training; and National Agricultural Extension and Research Liaison Services (NAERLS) in Nigeria drives awareness and farmer outreach. **In Ethiopia**, the Ministry of Agriculture and MILLS anchor policy and extension; Green Scene Energy (MoU with IWMI) leads market-based supply and scaling; and Lewegen MFI supports farmer finance. **In Kenya**, SDI (*MoU with IWMI*), NIA, and NEMA (*MoU with IWMI*) provide policy leadership; SunCulture leads solar technology supply; Strathmore University supports technical validation and capacity building; and GOGLA coordinates industry engagement, with KCB and Equity banks supporting farmer finance. The S4I team will work with partners through technical assistance on technology types, market segmentation, farmer organization, de-risking instruments, and co-designing loan products and scaling pathways.

- **Scaling Pathways**

The solution track will use a public-private partnership (PPP) approach to scale smallholder irrigation innovations, including solar irrigation, by combining stakeholder engagement, market linkage, capacity building, and policy engagement. The model recognizes that public sector investment alone cannot deliver the scale and service quality required for irrigation expansion; instead, it aligns government, private-sector suppliers, financiers, and farmer organizations to jointly deliver technology, financing, and irrigation services.

Scaling approaches differ slightly across the three countries:

- **Nigeria**: PPP approach to improve SPIS market linkage, business models (e.g., trailered business or IaaS), and facilitate financial access to farmers through de-risking (risk sharing mechanisms) agency like NIRSAL and a private bank (Serling Ltd).
- **Kenya**: PPP approach to mature the private sector market of SPSI through a finance facility, technical support (e.g., digital advisory), and capacity building
- **Ethiopia**: PPP approach to improve government coordination on SPSI, micro-finance access, technical support (e.g., digital advisory), and policy support

Scaling of primarily SPIS and other FLID bundles is underway in Nigeria (Kebbi, Kano, Jigawa and Adamawa states), Ethiopia (Rift Valley, central Ethiopia, and southern Ethiopia regions), and Kenya (Eastern Province, Rift Valley, and Western Province) due to strong demand, policy readiness, and growing markets for (solar-powered) FLID and ongoing public investment program. The last-mile delivery relies on private technology suppliers, financial institutions, beneficiaries of ToT, farmers' cooperatives, and public institutions.

S4I funding will be used to provide technical assistance, build partnerships, and deploy inclusive scaling pathways. This solution track receives support from various AoW3 PoDs, such as Accelerating Agribusiness for ImpactS4I in three countries, the Kenya Tariff and Subsidy Framework in Kenya, the Nigeria Enabling

Environment Scaling Toolkit, and Inclusive and Responsible Scaling in three countries. In Nigeria, IFPRI leads market linkages activties in Jigawa state, IWMI leads in Kebbi, and both FPRI and IWMI jointly work in Kano. This needs to account for country contexts (Kenya and Nigeria).

We chose this pathway because evidence shows that financial innovations with established partnerships among public and private approaches outperform traditional commercial finance in terms of durability and scale. As this approach allows flexibility, we can pivot between solar-powered and farmer-led irrigation by targeting individual farmers and cooperatives or farmers clusters to own SPIS or irrigation-as-a-service (as propsoed in Jigawa state in Nigeria), depending on local conditions.

Theory of Change (ToC): Our ToC is clear. If governments, financial institutions, private sector suppliers, and development partners are coordinated through formal and informal public-private partnership such as through national scaling platforms (eg in Ethiopia and Kenya), multi-stakeholder platforms (e.g., in Nigeria) and if farmers are supported with technical advisory services, appropriate financing mechanisms, and awareness of solar irrigation technologies, then barriers to adoption will be reduced, and investment in irrigation technologies will increase. Strengthened extension systems, improved supply chains, improved financial products, and supportive policies will enable farmers to adopt solar irrigation systems at scale, contributing to climate-resilient agricultural intensification, improved water management, and increased smallholder productivity across Ethiopia, Kenya, and Nigeria.

- **Opportunities and risks**

Opportunities

- The National Science and Engineering Infrastructure (NASEN) in Nigeria has a national initiative of reaching 50,000 farmers for SPIS. And Nigeria Incentive-based Risk Sharing System for Agricultural Lending (NIRSAL) has a national plan to de-risk investment for 10,000 farmers under SPIS.
- The initiatives of advancing solar-powered irrigation and liquified natural gas (LNG) pumps in the Jigawa state government of Nigeria, using PPP and irrigation as a service model.
- Leveraging on the on-going initiatives of Gates Foundation (GF) supported irrigation technology suitability mapping activities in Ethiopia, Nigeria, and Kenya
- SunCulture in Kenya has a target of distributing solar pumps to 20000 farmers in Kenya and 5000 farmers in Uganada,
- The opportunity to get support from different PoDs of AoW3, such as accelerating agribusiness for impact PoD (identifying investment ready pipelines), inclusive and responsible scaling PoD to support scaling partners in being inclusive and responsible, the water tariff PoD in Kenya to support the sustainable FLID investment, and understanding scaling barriers in Nigeria from the PoD of Nigeria Enabling Environment Scaling Toolkit,

Risks and Management

- Finance access and risk perception: FI may remain cautious in lending to smallholder irrigation due to perceived risks. Mitigation is to engage with de-risking institutions such as NAERLS in Nigeria and develop a sustainable solution, like blended finance in Kenya
- Fragmented supply chains: limited coordination between suppliers, financiers, and farmers may slow scaling. Mitigation involves establishing a scaling platform in Ethiopia and using the existing MSD platform in Nigeria.
- Institutional and policy barriers: regulatory gaps and weak policy enforcement may constrain scaling. Mitigation is to use the various PoDs under AoW3 to inform scaling partners to make incremental reforms.

- Risk of exclusion: women and youth may be excluded from the scaling. Mitigation involves engaging the inclusive and responsive scaling PoD with the scaling partners in the three countries through capacity building, inclusive adoption, and the scaling of innovation to mitigate unintended consequences.
- **Scaling track implementation plan (2026–2028)**

Table 2 presents a milestone-based scaling roadmap for 2026–2028, linking quarterly implementation actions to measurable evidence and adaptive decision points. It shows scaling milestones, decision points, and adaptive management needed, and the systematic collection of evidence. It links implementation actions to measurable outcomes and adaptive learning mechanisms, enabling timely monitoring, reflection, and course correction to ensure the solution track remains on course to achieve its ambitions.

Table 2. Milestone-based scaling roadmap for 2026–2028

2026				
Q1	Country pipelines validated	Confirm scaling and institutional partnership plans (workshops) Identify and agree on key technical/ institutional support needed with each partner South-south learning meetings (bilateral) Formation of scaling platform Discuss key learning questions	Updated agreed action plan (activities and timeline) Partner targets by country Baseline adoption data List of support types Policy landscape Finance diagnostics	Review partners' readiness and finance diagnostics. Validate priority regions and delivery partners Document risks and assumptions Refine country action plans and support
Q2	Scaling pathways operationalized by farmers, public institutions, and private partners through targeted capacity building and awareness programs	IWMI Training material identified/developed Capacity building through ToT to the extension system and private institutions Conduct demand-supply linkage Technical assistance to scaling partners on business models and finance facility IFPRI – Jigawa and Kano States: (1) Operationalize the PPP in Jigawa with the Ministry of Agriculture (MoA), Jigawa Agric and Rural Development Authority (JARDA), Jigawa Agricultural Transformation Agency (JATA), and farmers' clusters to facilitate the deployment of 1500 fuel-based irrigation pumps. (2) Facilitate irrigation as Service (IaaS) on solar pumps in collaboration with	Training completion records Pre/post knowledge assessments Scaling approaches Farmer uptake intentions Feedback for adaptive learning	Analyze pre- and post-training and partner feedback Identify gaps in institutional and farmers' understanding of scaling pathways Adapt training content, delivery methods, and awareness strategies Adjust partnerships and PPP arrangements

		JARDA, JATA, and Private sector. (3) Facilitate the PPP model in Kano working with ADUST and Gaya traditional authority (4) Training of extension agents (in collaboration with Jigawa MoA and Kano MoA)		
Q3	Early scaling proof achieved	Training and awareness through the media houses Capacity building through ToT to the extension system and private institutions Monitor, learn, and adapt demand-supply linkages (IFPRI) Technical assistance to scaling partners on business models and (micro) finance facility in collaboration with JARDA, NIRSAL and Sterling Bank Ltd. (IFPRI) Disseminate/train the SSI decision support tools to intermediary actors (government, private sector, and financiers) in Jigawa and Kano (IFPRI) Digital tool training Rollout digital advisory Mid-year reflection workshop (½ day adaptive learning)	Adoption and awareness rates ≥ 40% of targeted farmers. Irrigated areas data Gender disaggregated participation rates Youth participation	Review mid-year performance data Assess the effectiveness of the finance mechanisms and advisory services Identify bottlenecks and successful practices Rebalance delivery
Q4	Expand outreach, strengthen partners' engagement, and learn from policy lessons	Public-private partnership (PPP) to support the distribution channel Policy briefs (IFPRI lead) Annual learning workshop (adaptive	Adoption rate reach 100% of targeted farmers Institutional readiness scores	Synthesize annual learning Evaluate partners' performance Integrate lessons into revised scaling

		learning) Plan for 2027	Finance transactions	pathways Confirm 2027 priorities and resource allocations
2027				
Q1-2	Finance and delivery systems scaled	New regions identified Expand the technical and extension support through training and capacity building Scaling finance facilities and access based on PoDs of AoW3 Private supplier expansion New PPP establishments	Adoption rates reach ≥ 40% of targeted farmers. Training completion New partners' growth data, Institutional readiness, Cost reductions	Review performance Assess institutional capacity Adjust risk-sharing mechanisms Prioritize high-performing partners and regions
Q3-4	2027 scaling targets verified, finance facility establishment in two countries, and regional replication initiated	New region entry in each country Re-define the scaling pathways Irrigation as a Service model for wider implementation Cross-country learning	Adoption rates reach 100% of targeted farmers Finance transaction Comparative performance data Service viability	Compare performance across countries Evaluate finance facility viability Identify dominant pathway
2028				
Q1-2	System-level integration and implementation of scaling	Embed in national programs Budget alignment Regulatory support	Adoption rates reach ≥ 40% of targeted farmers Policy uptake evidence Public finance leverage Verified farmers reached	Assess alignment with national policies, budgets, and regulatory framework
Q3-4	Sustainable scaling consolidated	Private-led expansion Exit from intensive support Final impact review	Adoption rate reach 100% of targeted farmers Long-term	Conduct final impact assessment Evaluate

	adoption trends Environmental indicators Investment leverage	transferable lessons

- **Complementary bilateral projects and S4I additionality**

S4I resources strategically leverage ongoing bilateral initiatives to enhance scalability, coordination, and long-term sustainability. The SoLAR Phase II project provides systems-level support for solar agriculture in Ethiopia and Kenya, NSSID strengthens piloting and policy integration in Ethiopia, and the IFPRI-led Irrigation Investment Tool informs investment decisions in Nigeria. Building on these platforms, S4I adds value by consolidating 2025 outcomes and bundling solar and farmer-led irrigation technologies with inclusive finance, digital advisory tools (i.e., developed through bilateral projects), and institutional capacity development. S4I funding supports key intermediaries that link farmers to financial institutions, reducing risk and improving access to credit.